Parkinson's Classification

Stefano Colella

865075

Introduction

I present a binary classification pipeline to distinguish

Parkinson's disease (PD) patients (1)
from healthy subjects (0)

using the PaHaW Handwriting Dataset .

Our goal is to extract reliable indicators of PD from time -series tablet data .

STATE OF THE ART

The dataset contains time-series handwriting samples :

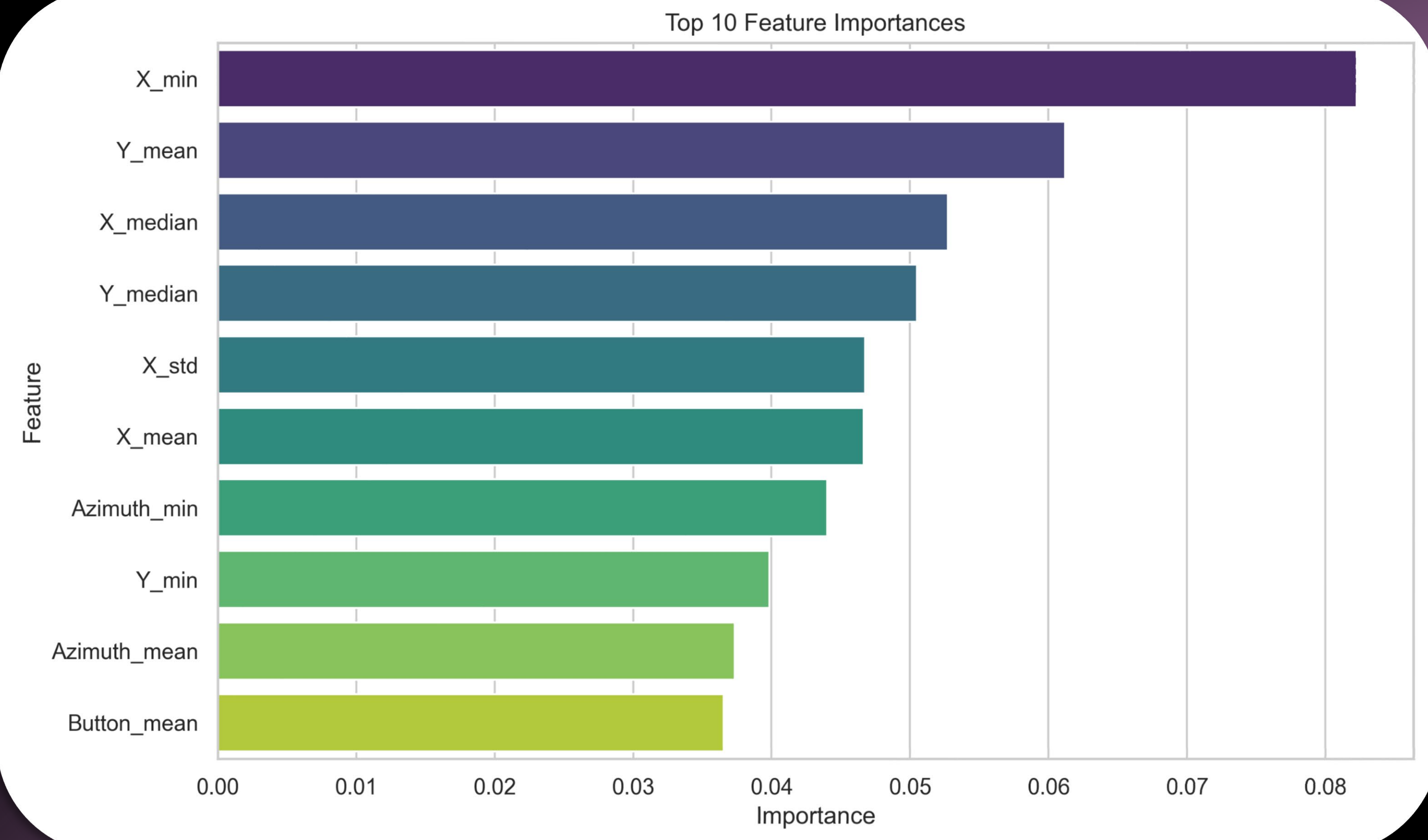
- X/Y coordinates
- pen pressure
- Altitude
- Azimuth

from 72 subjects (37 with Parkinson Diagnosis and 38 without) .

ID	Natio...	Sex	Disease	PD stat...	Age	Dominant hand	LED	UPDRS V	Length of PD
00001	Czech	F	PD	ON	68	R	1115	2	6
00002	Czech	F	PD	ON	78	R	2110	2	8
00003	Czech	F	PD	ON	69	R	1556.6	2	7
00004	Czech	F	PD	ON	79	R	1691	2	12
00009	Czech	M	PD	ON	78	R	2066	1	3
00010	Czech	M	PD	ON	74	R	1480	2.5	3

Subjects 00061, 00080, and 00089 were excluded due to incomplete data.

Materials & Dataset



Raw time-series files (.svc) were processed to extract static and dynamic features.

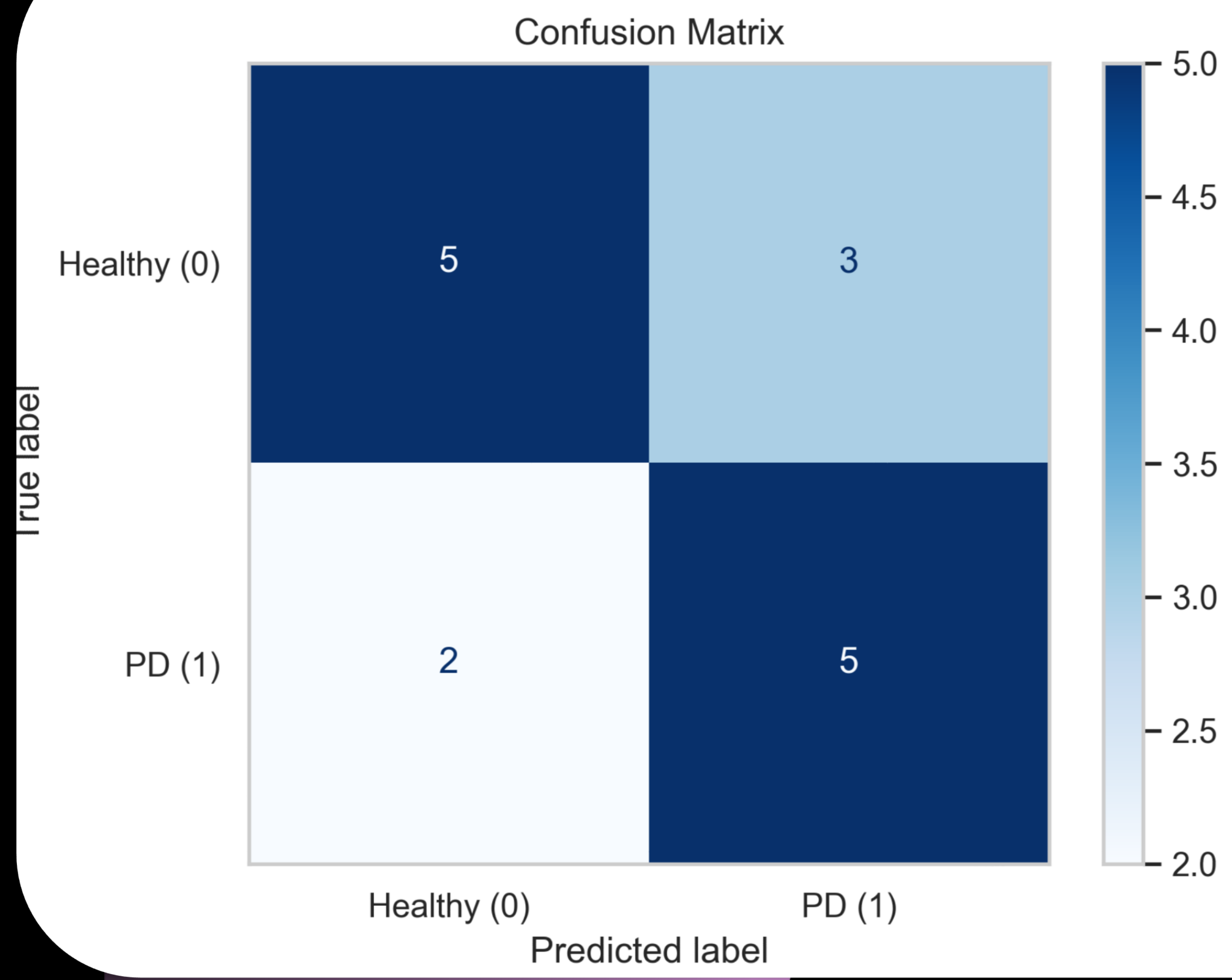
We calculated continuous velocity from spatial displacements and time deltas. We computed the mean, std, max, min, and median for all physical properties.

Model Training & features engineering

We deployed a Random Forest Classifier. To prevent overfitting on small data, hyperparameters were optimized:

- n_estimators =200
- max_depth =7
- min_samples_split =5
- min_samples_leaf =2
- max_features ='sqrt'
- class_weight ='balanced'

Future improvements could involve employing Convolutional Neural Networks on Spectrogram representations of the time series to capture more intricate temporal dependencies .



EVALUATION METRICS

Conclusion

Precision: 0.625

Specificity: 0.625

F1-Score: 0.6667

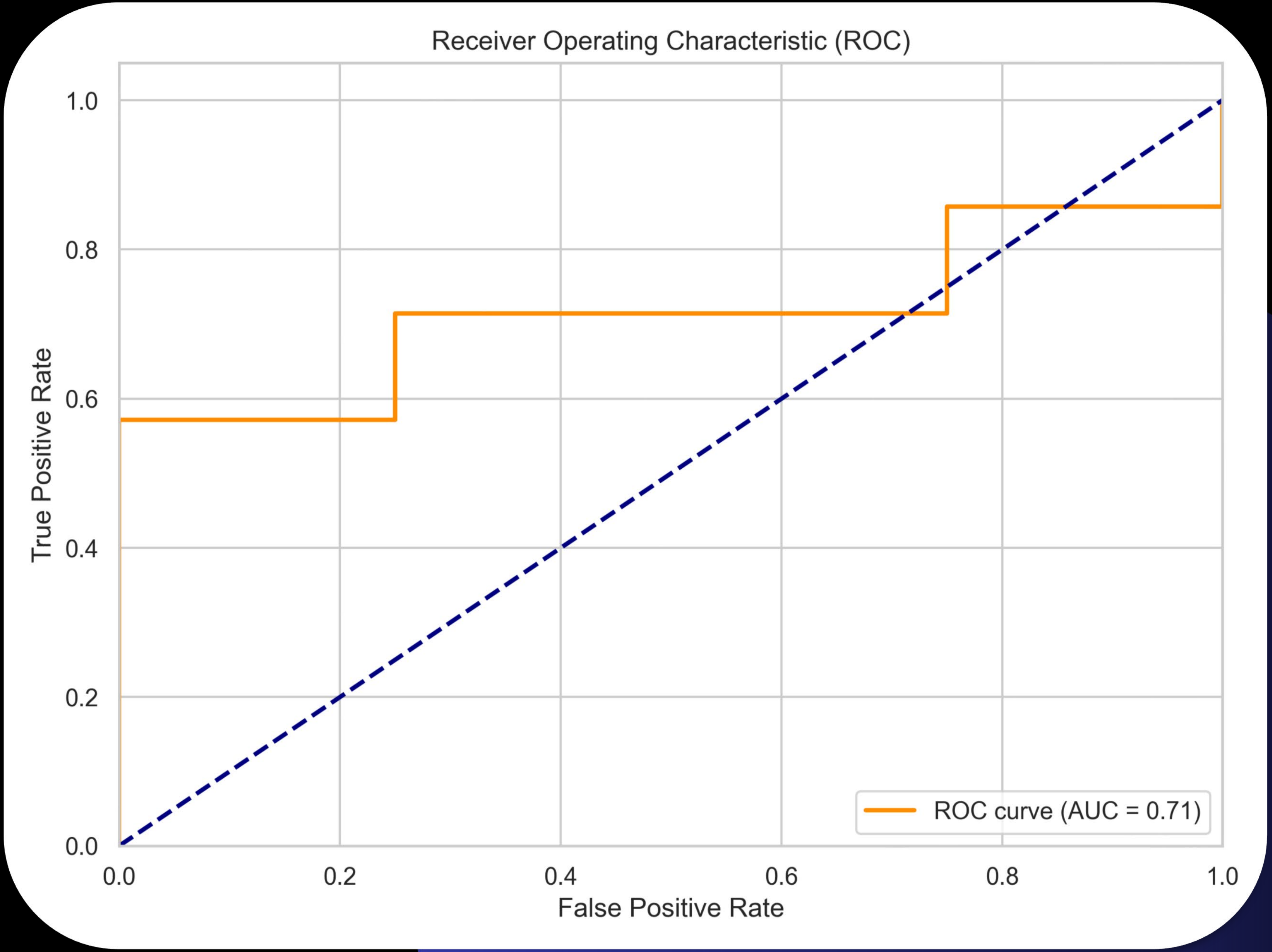
Accuracy: 0.6667

Recall: 0.7143

AUC: 0.7143

The model demonstrates a strong capability to identify PD patients

The Random Forest naturally resisted overfitting, establishing a highly competitive baseline .



Future improvements could involve employing Convolutional Neural Networks on Spectrogram representations of the time series to capture more intricate temporal dependencies .

Future Work